week-11-overview

Week 11 - Overview
Welcome to week 11

Spark Streaming is an extension of the core Spark API that enables scalable, high-throughput, fault-tolerant stream processing of live data streams. Data can be ingested from many sources like Kafka, Kinesis, or TCP sockets, and can be processed using complex algorithms expressed with high-level functions like map, reduce, join and window. Finally, processed data can be pushed out to filesystems, databases, and live dashboards. In fact, you can apply Spark's machine learning and graph processing algorithms on data streams.

Learning Outcomes

CLO 3: motivate and explain trade-offs in big data processing technique design and analysis in written and oral form.
CLO 4: explain the Big Data Fundamentals, including the evolution of Big Data, the characteristics of Big Data and the challenges introduced.
CLO 5: apply non-relational databases, the techniques for storing and processing large volumes of structured and unstructured data, as well as streaming data.
CLO 6: apply the novel architectures and platforms introduced for Big data, i.e., Hadoop, MapReduce, and Spark.
This Week's Learning Activities

Section Activity Duration

Before Class

Watch the video lectures and

Complete reading

2 hours

Lectorial & Lab

Join the timetabled interactive session

2 hours + 2 hours

Assessments

Working on Assessment

Revisit lectures and complete the Weekly Quiz

2 hours + 2 hours

Previous
Next

LINKS:
- Previous: https://rmit.instructure.com/courses/158408/modules/items/8688614
- Next: https://rmit.instructure.com/courses/158408/modules/items/8688616